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BRIEF REVIEW

The Effect of Training Loads on Performance Measures and Injury Characteristics in Rugby League Players: A Systematic Review

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Purpose: To conduct a systematic review into the effect of training load (TL) on both performance measures and injury characteristics in rugby league players. ***Methods:*** Based on PRISMA guidelines, a systematic search of electronic databases was performed from the earliest record to December 2016. Five electronic databases (MEDLINE, SPORTDiscus, CINAHL, Web of Science, and AusSportMed) were searched using keywords within the subcategories of population (rugby league players), intervention (training quantification), and outcomes (performance or injury). Original peer-reviewed published manuscripts were considered if they had a clear quantification of TL (eg, session rating of perceived exertion), clear measures of performance (tests of physiological parameters relative to rugby league players [eg, agility]), and/or injury characteristics (eg, injury incidence). ***Results:*** Twelve studies met the inclusion criteria. The highest TL was observed in preseason training phases. Higher TL elicited greater improvements in maximal aerobic power, vertical jump (in cm), and agility scores (in s). Higher TL was associated with higher incidence of muscular strains and joint sprains, lower-limb injuries, and overexertion and overuse injuries. ***Conclusions:*** Findings suggest a dose–response relationship between TL, physical performance, and injury characteristics. However, due to study and methodological limitations, the precise nature of this relationship is, as yet, inconclusive. Therefore, the development of uniform TL definition and quantification and further research is warranted.

Keywords: injury management, physical performance, sport, athletic training

Elite athlete development necessitates years of structured, progressive, and intense training,[1] incorporating a multidisciplinary approach including skill acquisition, strength and conditioning, and psychological development. In particular, high physical training loads (TLs) over extended time periods are required to provide adequate stimulus to attain the desired performance enhancements.[2] However, high progressive TL can be accompanied by increased exposure to negative outcomes, such as nonfunctional overreaching,[3] injury,[4] illness,[5] and burnout.[6] At the same time, lower levels of aerobic capacity and muscular power have also been attributed to increased injury incidence.[7,8] As competitive level increases, the training dose increases to elicit adaptations that match the requirements of higher performance levels.[9] Thus, it is crucial that TLs are carefully monitored to maximize positive training adaptations (eg, increased aerobic power, muscular power, and agility) while minimizing exposure to negative outcomes (eg, overreaching, injury, and illness).

Accurate assessment of the relationships between TL, performance enhancement, and injury would allow more optimal prescription of targeted and periodized training programs to develop physical capacities while simultaneously minimizing negative outcomes. Recent research developments such as the acute:chronic workload ratio have led to enhancements in the application and monitoring of TL in training programs. However, as yet there is inadequate evidence on which to facilitate major progress in training practices based on quantification of TL.

Training load assessment is particularly valuable in team-sport contexts such as rugby league, a collision sport involving frequent high-intensity efforts (eg, sprinting and tackling) punctuated by short bouts of low-intensity activity (ie, walking).[10] It is one of the most injurious Australian sports for players aged 5–14 years (31.4 injuries/100,000 persons) and 15–34 years (55.6 injuries/100,000 persons).[11,12] Around 4.9% of sporting injury hospitalizations in Australia in 2011–2012 occurred due to rugby league participation.[13] Musculoskeletal injuries,[14] categorized as contact or noncontact injuries, are common. High incidences of soft-tissue injury have been previously observed in training.[15] Relationships between overall TL and injury characteristics such as injury type,[16] site of injury,[16,17] mechanism of injury,[17] and injury severity[17,18] have been examined.

Various methods (internal and/or external) have been used to quantify TL in existing studies. Internal measures include ratings of perceived exertion (RPE) or modified RPE (Borg scale 1–10),[19] session rating of perceived exertion (sRPE) in arbitrary units (AU) calculated by RPE × session duration,[20] training impulse method,[21] heart rate variation,[22] or a combinations of such methods.[23,24] By contrast, external measures of TL used thus far include global positioning systems (GPS) with microtechnology sensors (measures of distance [in m], speed [in $m \cdot s^{-1}$], acceleration/deceleration [in $m \cdot s^{-2}$]).[25] To date, rugby league studies have used a modified RPE[26]; sRPE reported as AU[15]; and GPS variables, particularly total and high-speed distances (in m),[27] to quantify TL.

Current literature in rowing,[28] tennis,[29] soccer,[30] basketball,[31] swimming,[32] and running[33] indicates relationships exist between TL, physical performance, and injury characteristics.[15,16,34] However, at present the existing body of empirical evidence examining the relationship between TL, performance enhancement, and injury characteristics in rugby league is limited.[16,17,34–36] Such limitations include a small number of independent studies in total, methodological limitations in the existing literature (eg, lack of standardized quantification and reporting of TL, issues of external validity and bias [incomplete reporting of recruitment procedures and compliance] and largely homologous study populations, and inconsistent reporting of injury characteristics).

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Thus, with these issues in mind, the purpose of this study was to conduct a systematic review and investigate the effect of TL on physical performance and injury characteristics in rugby league.

Methods

Search Strategy

The systematic review process was conducted in accordance with PRISMA guidelines.[37] A systematic search of electronic databases (MEDLINE, SPORTDiscus, CINAHL, Web of Science, and AusSportMed) was performed from the earliest record to December 2016. The search strategy was formulated under the subcategories of population, intervention, and outcomes. These categories were chosen to define a specific set of subjects, intervention type (TL was quantified), and precise measurable outcomes (eg, injury characteristics and/or performance measures). Population search terms were the sport of rugby league and associated terms ("rugby*," OR "rugby league," OR "rugby league player*"). Intervention terms were all associated with training quantification and included "training load," OR "training volume," OR "volume load," OR "RPE," OR "sRPE" OR "iTRIMP," OR "resistance training," OR "strength training," OR "physical preparation," OR "GPS," OR "global positioning system." Outcome terms attempted to capture areas of performance or injury ("physical performance," OR "hypertrophy," OR "sport performance," OR "strength," OR "1RM" OR "3RM," OR "5RM," OR "bench throw" OR "jump squat," OR "vertical jump" OR "CMJ," OR "jump height," OR "sprint," OR "running velocity," OR "acceleration," OR "maximal speed," OR "change of direction," OR "COD," OR "agility" OR "505," OR "injury," OR "strain," OR "sprain," OR "contusion," OR "laceration" OR "concussion," OR "soft tissue," OR "contact," OR "non-contact," OR "haematoma"). Results of these 3 searches were then combined using the Boolean phrase AND.

Study Eligibility

All study design types were included. Only original peer-reviewed published manuscripts were considered. Studies were included if they

- Included male or female rugby league players at any age or competitive level
- Defined a clear quantification of TL (eg, sRPE, GPS measures) for either field sessions (conditioning/skills) or resistance training
- Defined quantifiable measures of performance (eg, strength, power, maximal aerobic power, speed, and agility) performed both before and after training interventions
- Recorded and analyzed injury characteristics (eg, number of injuries, injury incidence rate, injury location, injury severity, mechanism of injury)

Studies were excluded if they

- Were review papers, abstracts, or conference papers from annual meetings.
- Were not exclusively conducted on rugby league players.
- Did not define a clear quantification of TL
- Did not include either a quantifiable preperformance or postperformance measure, or injury characteristics had not been recorded and analyzed

After eliminating duplicates, search results were screened independently by 2 researchers against eligibility criteria. References that could not be eliminated by title or abstract were retrieved and independently evaluated. Reviewers were not masked to title or authors of identified publications. Disagreements were resolved by discussion or via a third researcher. Reference lists of all retrieved studies were manually searched for other potentially eligible studies. The iterative reference check included specific search terms matching those used in the database search, such as rugby league, TL, injury incidence and/or characteristics, and performance measures (eg, sprint, aerobic power, muscular power).

Methodological Quality Assessment

The quality of included studies was independently assessed by 2 researchers using the Downs and Black methodological assessment tool (Table 1).[41] The items used reflected the quality subscales of reporting (numbers 1–12), external validity (13–17), internal validity bias (18–25), and confounding (26–33).

Data Extraction

Data relating to participant characteristics (ie, age, height, body mass, and level of competition); training activity (modality, TL); and performance variables (maximal aerobic power, maximal running speed, muscular power, and agility) as well as training injury data were independently extracted by 2 researchers.

Training load was recorded as sRPE (RPE multiplied by session duration [in min]) and reported in AU. Here, 6 (50%) of the studies included were conducted prior to the introduction of GPS technology in rugby league and another 3 (25%) were conducted before its application was commonplace in the sport. The remaining 3 more recent papers all used GPS data to quantify TL variables, which were categorized according to total distance (in m), relative distance covered (in $m \cdot min^{-1}$), or total high-intensity running distance (in m).

Performance measures were recorded as reported in all studies. The terms consistently defined across all studies investigating performance were 10-, 20-, and 40-m sprint times (in s) and maximum sprint velocity (in $m \cdot s^{-1}$). Agility measures were reported as actual times recorded (in s). Lower body muscular power was estimated from the vertical jump test using highest jump height (in cm). Maximal aerobic power was reported as predicted maximum in relative measures (VO_2max, in $mL \cdot kg^{-1} \cdot min^{-1}$).

Recently, the World Health Organization has presented the notion of "sports impairment," which has led to more comprehensive definitions of injury. For example, "sports injury" denotes the loss of bodily function or structure that is the object of observations in clinical examinations; "sports trauma" is defined as an immediate sensation of pain, discomfort, or loss of functioning that is the object of athlete self-evaluations; and "sports incapacity" is the sidelining of an athlete because of a health evaluation made by a legitimate sports authority that is the object of time loss observations.[42] The World Health Organization definition of sports "injury" and "incapacity" align well with the definition provided by the studies included in this review. Studies included defined an "injury" as any pain or disability that occurred during training or match play as assessed by the club head trainer either during or after training or match play.[26] An injury was recorded if it resulted in either a stoppage in training or play, or caused transient or permanent incapacity. Training injury exposure was calculated by multiplying the number of players by the number and duration

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Table 1 Downs and Black Methodological Assessment

Downs and Black criteria	Hulin et al[38]	Hulin et al[39]	Gabbett and Ullah[27]	Gabbett and Jenkins[18]	Gabbett[26]	Killen et al[40]	Gabbett and Domrow[16]	Gabbett[34]	Gabbett[7,35,36]	Gabbett[7,35,36]	Gabbett[15,17]	Gabbett[15,17]
Reporting												
1. Hypothesis/aim clearly stated	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
2. Underlying theory described	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3. Main outcomes described in intro or method	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
4. Subject characteristics described	Yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No
5. Intervention described	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
6. Main exposure to intervention measured	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
7. Distribution of confounders clearly described	No	No	No	No	No	No	No	No	No	No	No	No
8. Main findings described	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
9. Estimate of random variability for main outcome	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
10. All adverse events reported	No	No	No	No	No	No	No	No	No	No	No	No
11. Participants lost to follow-up described	No	No	No	No	No	No	No	No	No	No	No	No
12. Actual probability value reported	No	No	No	No	No	No	Yes	No	No	No	No	No
Reporting score	8	8	7	8	8	7	9	8	8	8	7	7
Validity												
13. Potential recruits representative of entire population	U	U	U	U	U	U	U	U	U	U	U	U
14. Participants representative of entire population from which they were recruited	No	No	No	No	No	No	No	Yes	No	No	Yes	No
15. Staff and facilities representative of the intervention	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
16. Screening criteria defined	No	No	No	No	No	No	No	No	No	No	No	No
17. Attempt to blind participants	No	No	No	No	No	No	No	No	No	No	No	No
Validity score	1	1	1	1	1	1	1	2	1	1	2	1
Bias												
18. Attempt to blind assessors	No	No	No	No	No	No	No	No	No	No	No	No
19. Appropriate methods to adjust for differences between groups	No	No	No	No	No	No	No	No	U	Yes	No	No
20. Appropriate methods used to account for biases related to differential outcomes between groups	No	No	No	No	No	No	No	No	U	Yes	No	No
21. Was any data dredging identified	U	U	U	U	U	U	U	U	U	U	U	U

(continued)

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Table 1 *(continued)*

Downs and Black criteria	Hulin et al[38]	Hulin et al[39]	Gabbett and Ullah[27]	Gabbett and Jenkins[18]	Gabbett[26]	Killen et al[40]	Gabbett and Domrow[16]	Gabbett[34]	Gabbett[7,35,36]	Gabbett[7,35,36]	Gabbett[15,17]	Gabbett[15,17]
22. Did analysis adjust for differences in length of time to follow-up between groups	No	No	No	No	No	No	No	No	U	U	No	No
23. Statistical tests appropriate	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
24. Compliance reliable	U	U	U	U	U	U	U	U	U	U	U	U
25. Outcome measures valid and reliable	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bias score	2	2	2	2	2	2	2	2	2	4	2	2
Confounding												
26. Were participants from different groups from same population	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
27. Were participants from different groups recruited at the same time	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
28. Were participants randomized	No	No	No	No	No	No	No	No	No	No	No	No
29. Was randomization concealed	U	U	U	U	U	U	U	U	U	U	U	U
30. Were participants aware of hypothesis	U	U	U	U	U	U	U	U	U	U	U	U
31. Was there adequate adjustment for confounding	No	No	Yes	No	No	No	No	No	No	No	No	No
32. Losses to follow-up accounted for	U	U	U	U	U	U	U	U	U	U	U	U
33. Power analysis conducted	No	No	No	No	No	No	No	No	No	No	No	No
Confounding score	2	2	3	2	2	2	2	2	2	2	2	2
Total score	13	13	13	13	13	12	14	14	13	15	13	12

Abbreviation: U, unable to determine.

of training sessions.[43] Injury incidence was calculated as number of injuries/training injury exposure per 1000 training hours (TH).[43] Injury data were also reported according to the following categories: injury location (body part); injury severity (transient [no game or training time lost], minor [up to 7 d missed or 1 game missed], moderate [8–28 d or 2–4 matches missed], and major [>28 d or >4 games missed]; and mechanism of injury (eg, player contact, while tackling). Relative risk (RR) for a given threshold for GPS variables (eg, RR for discrete movement velocities, discrete acceleration bands, and repeated high-intensity efforts >3) was also recorded.

Results

Identification of Studies

A total of 1316 studies were retrieved from 5 electronic databases (SPORTDiscus [N = 473], CINAHL [N = 189], Web of Science [N = 306], MEDLINE [N = 334], and AusSportMed [N = 14]) and 11 via hand searches of bibliographies. Duplicates (N = 913) were removed. After screening titles and abstracts, 12 studies met the inclusion criteria (Figure 1). Five of these examined the relationship between TL and performance as well as injury incidence,[15,16,34–36] while all 12 included studies investigated the relationship between TL and injury incidence.

Study Characteristics

Methodological quality scores were assessed using the Downs and Black measurement tool (Table 1). Studies scored highest for the reporting domain (7–9/12)—specifically, in definitions of populations, exposure, and outcome measures as well as between-group statistical comparisons and measures of variability. External (1–2/5) and internal validity (2–4/7) measures and confounding (2–3/8) all scored lower due to a lack of control group, blinding, or randomization. Furthermore, discussion of participant dropout and follow-up, along with power calculations for sample sizes, were also lacking in all studies. Total Downs and Black scores ranged between 12 and 15/33.

A total of 965 rugby league participants across all competition levels were reported in the 12 identified studies: National Rugby League (n = 293)[18,26,27,38,40]; Queensland Cup (n = 109)[34,36]; semi-professional (n = 440),[15–17,34,36] amateur (n = 22),[15] under 19 years (n = 20),[15] and juniors aged 14–18 years (n = 81).[34,35] Mean reported age ranges for studies investigating both performance measures and injury were 16.9 (0.2)–25.5 (5.1) years. All 12

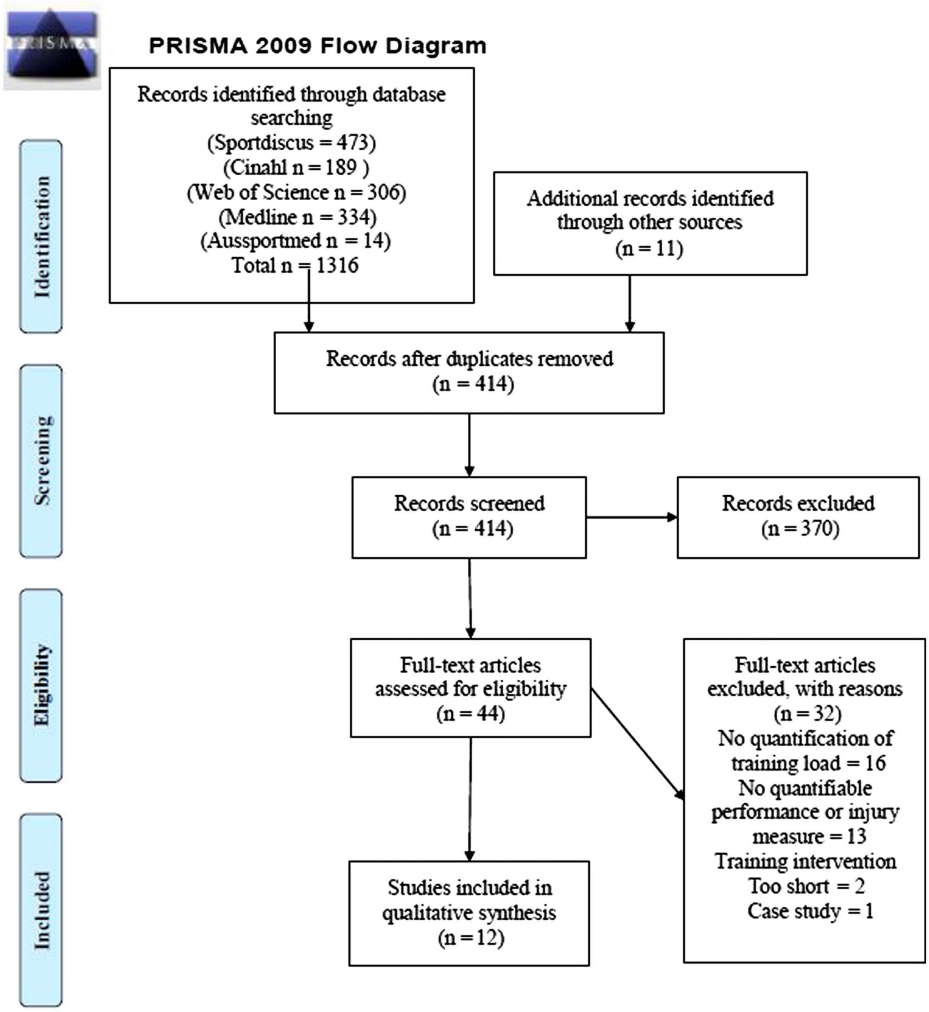

Figure 1 — PRISMA 2009 flow diagram (from Moher et al[44]).

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studies (100%) reported on Australian male (predominantly Queensland) players. No studies on female athletes fitting the search criteria were identified.

Five studies (42%) investigated the effects of TLs on the performance measures[16,17,34–36] of aerobic power, vertical jump height, and running speed (10-, 20-, 40-m sprints). Four studies (33%) reported the effects of TL on agility (L run).[15,16,35,36] Six studies (50%) characterized injury type (eg, muscular strain) and site.[15–18,26,40] Four studies (33%) stated injury mechanisms (eg, being tackled).[15–18] Five studies (50%) stated forms of injury cost (eg, time lost in training or matches).[15,17,18,27,40] Two studies (16.7%) compared different playing levels and/or age groups on injury incidence.[15,34] Two studies (16.7%) focused solely on soft-tissue injuries during training.[26,27] One study (8.3%) divided injury into contact, noncontact, and strength (sustained in the gym) injuries.[18] Four studies (33.3%) assessed training injuries longitudinally over consecutive seasons, covering either 2, 3, or 4 years.[16–18,26] All studies reported injury data as total number and incidence.

Nine studies (75%) used sRPE reported as AU to determine training session TL.[15–18,26,34–36,40] The remaining 3 (25%) studies used GPS technology (total distance).[27,38,39] One of these reported further on measures of relative distance (in meters per minute); discrete movement velocities (very low [0–1 $m \cdot s^{-1}$], low [1–3 $m \cdot s^{-1}$], moderate [3–5 $m \cdot s^{-1}$], high [5–7 $m \cdot s^{-1}$], and very high [>7 $m \cdot s^{-1}$]); discrete acceleration bands (mild [0.55–1.11 $m \cdot s^{-2}$], moderate [1.12–2.78 $m \cdot s^{-2}$], and maximal [>2.79 $m \cdot s^{-2}$]); and repeated high-intensity efforts. Only 1 study (8.7%) quantified resistance TL by using sRPE × duration (AU).[18]

Training frequencies in rugby league were reported at between 2 field sessions per week for junior athletes[34,35] up to as many as 5 field sessions and 4 structured gym sessions for elite (professional rugby league players) athletes in the preseason training phase.[18] The duration of training sessions was typically between 60 and 100 min.[16,34] Training intensities assessed via sRPE in all studies ranged from 3.3 in early competition to 4.7 in early preseason.[17] Overall TL ranged from 155 to 620 AU per training session.[15–17,36] Training intensities[36] and average TL[36] were shown to increase from the beginning of preseason to the end of preseason, then to decrease at the end of the competition phase.

TL and Physical Performance

Five studies investigated the effect of TL on measures of physical performance (Table 2).[16,17,34–36] Improvements in aerobic power were observed in all studies in the training groups, ranging from 2.3 $mL \cdot kg^{-1} \cdot min^{-1}$ (5.1%; average session TL = 469 AU) in semiprofessional players[17] to 9.3 $mL \cdot kg^{-1} \cdot min^{-1}$ (22%; average session TL = 202–275 AU) in elite juniors (representative level).[36]

Improvements in muscular power (vertical jump) were observed in all but 1 study.[36] Increases between 0.2 cm (0.003%; average session TL = 326 AU) in semiprofessional players and 4.2 cm (7.8%; average session TL = 355.9 AU) in junior elite players were observed. Intraseasonal improvements were observed from the beginning of preseason to precompetition measures in 3 studies; however, these improvements were only retained throughout the season in 2 studies.[16,35]

Maximal speed was assessed in 5 studies.[16,17,34–36] Three studies[34–36] in elite junior and subelite senior players showed no changes in maximal speed measures over any distance from pre to post. By contrast, a statistically significant 1.6% improvement in maximal velocity over 20 m was observed from the beginning of preseason to precompetition (1891 AU weekly), which was maintained at midseason and postseason (1416.5 AU weekly).[16]

Agility was assessed in 4 studies.[16,34–36] Reductions in "L Run" times were seen in all studies, ranging from −0.09 seconds (1.5%; average session TL = 275.9 AU) to −1.03 seconds (17.8%; average session TL = 355 AU) from preseason to postseason. Junior athletes exhibited marginally greater reductions (−1.03 s, 17.8%) than senior athletes (−0.97 s, 16.2%).[34] Improvements in agility scores were seen in 3 studies[16,35,36] from off-season to precompetition measures; however, these improvements were only maintained throughout the season in 2 studies.[35,36] The remaining study in semiprofessional players[16] showed a return to near baseline values at the midseason and postseason tests.

TL and Injury Characteristics

Relationships between TL and overall injury incidence were apparent in 6 studies.[15–18,27,38] The remaining 6 studies either did not report specific TL or total incidence, or demonstrate a relationship between the two. The nature of the relationships identified varied depending on competitive level,[15,34] study duration,[17,40] and method of TL measurement.[27] Higher competition levels were associated with higher injury incidence from semiprofessional (243.2 AU) to under 19s (209.7 AU; $P < .001$).[15] In 1 study, higher injury incidence was observed over 4 months in the 2001 preseason period (156.7/1000 TH, average session TL = 278 AU) than the 2002 preseason (94.4/1000 TH, average session TL = 237 AU) and 2003 (78.4/1000 TH, average session TL = 221 AU).[17] These increases in injury incidence (105.2–205.5/1000 TH and 53.3–114.9/1000 TH) were also observed intraseasonally from early to late preseason training in 2001 and 2003 with concurrent increases in TL (278–385 AU and 221–336 AU) in subelite players.[17]

When examined over an entire season (December to September), preseason (December to February) overall injury incidence (137.7/1000 TH) was significantly higher than early and late season incidences (76 and 62.6/1000 TH, respectively) ($P = .001$). These training injury incidence rates declined in accordance with decreasing TL (1891, 1345, 1488 AU, respectively; $P = .003$).[16] Similar observations were noted in elite-level players ($r = .82$, $P < .01$),[18] using external measures of TL (GPS). Furthermore, very high-velocity running (>7 $m \cdot s^{-1}$) was associated with a 2.7 times higher risk of transient soft-tissue injury when it exceeded a total of 9 m per session as calculated over an entire season.[27]

Six studies investigated the relationship between TL and injury type (Table 3),[15,16–18,26,40] with 2 identifying relationships. Muscular strains displayed the highest incidences of injury type regardless of TL, ranging from 1.3 to 160.6 per 1000 TH.[15–17,40] Higher incidences of all injury types were reported in semiprofessional players during preseason training in conjunction with higher weekly TL (1891 AU) compared with early (1345 AU weekly) or late competition training phases (1488 AU) ($P = .001$).[16] Higher TL during a 14-week preseason, across 3 seasons (326, 260, 298 AU average session TL 2001, 2002, 2003, respectively) ($P < .001$), was associated with significantly higher incidences of muscular strains (62.4/1000 TH, 23.2/1000 TH, 19.6/1000 TH) and joint sprains (43.7/1000 TH, 14.6/1000 TH, 14.9/1000 TH) ($P < .001$).[17]

Six studies investigated the relationship between TL and injury site (Table 4),[15,16–18,26,40] with 2 identifying relationships. The lower limb represented the highest proportion of injuries by site.[15–18,38,40] Thigh, calf, ankle, knee, and foot injuries exhibited significantly higher injury incidences than other regions. Increased

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Table 2 Summary of the Relationship Between Training Loads and Performance Variables in Rugby League Players

Study	Group	Reporting variable	Training load, AU	Maximal aerobic power; VO_2max, mL·kg^{-1}·min^{-1}	Vertical jump height, cm	Sprint speed, s			Agility
						10 m	20 m	40 m	L Run, s
Gabbett and Domrow[16]	Senior	Off-season	NA	43.4	53.8	4.97[a]	7.52[a]	8.08[a]	5.97
	Senior	Preseason	1891.0	47.8*	57.4**	4.99[a]	7.64[a,†]	8.05[a]	5.42*
	Senior	Midseason	1416.5	47.6*	57.3**	5.17[a,***]	7.56[a,†]	8.06[a]	5.96
	Senior	Postseason	NA	48.6*	57.3**	5.07[a,***]	7.59[a,†]	8.09[a]	5.97
Gabbett[34]	Junior	Preseason	355.9	46.3	54.4	1.80	3.08	5.48	5.81
		Postseason		50.3‡	58.6‡	1.84	3.09	5.54	4.78‡
	Senior	Preseason	469.7	44.9	57.3	1.85	3.16	5.63	5.96
		Postseason		47.2‡	60‡	1.85	3.11	5.59	4.99‡
Gabbett[35]	Training	Off-season	NA	43.7	54.8	1.82	3.12	5.56	5.93
		Preseason	275.9	50.6‡	58.2‡	1.85	3.11	5.58	5.84‡
		Midseason	202.7	53.5‡	56.8	1.82	3.13	5.64	5.55‡
		Postseason	NA	52.1‡	57.8‡	1.79	3.09	5.52	5.82‡
	Control	Off-season	NA	46.1	53.6	1.81	3.10	5.51	5.83
		Preseason		47.0	54.6	1.80	3.08	5.48	5.70
		Midseason		45.7	53.1	1.79	3.08	5.51	5.78
		Postseason		46.7	54.3	1.80	3.09	5.49	5.75
Gabbett[36]	Training	Off-season	NA	42.0	55.4	1.83	3.15	5.61	6.03
		Preseason	330.8	48.5	58.6	1.85	3.12	5.61	5.92
		Midseason	329.7	51.3	55.7	1.80	3.13	5.62	5.95
		Postseason	NA	49.6	55.5	1.80	3.10	5.53	5.94
	Control	Off-season	NA	44.9	53.4	1.86	3.17	5.62	5.96
		Preseason		45.4	54.0	1.85	3.16	5.61	5.89
		Midseason		43.1	51.7	1.81	3.11	5.56	5.92
		Postseason		45.3	53.8	1.85	3.16	5.61	5.91
Gabbett[15,17]	2001	Preseason	326.0	43.8	53.1	2.24	3.55	6.09	NR
		Postseason		47.2‡	53.3	2.22	3.54	6.05	
	2002	Preseason	260.0	40.7	54.6	2.15	3.49	5.98	NR
		Postseason		45.5‡	54.2	2.18	3.5	5.98	
	2003	Preseason	298.0	42.0	55.4	1.83	3.15	5.61	NR
		Postseason		48.5‡	58.6	1.85	3.12	5.61	

Abbreviations: AU, arbitrary units; NA, not applicable; NR, not reported; VO_2max, maximal oxygen uptake.
[a] Reported as peak velocity m·s^{-1}.
*Significantly different from baseline ($P = .001$). **Significantly different from baseline ($P = .04$). ***Significantly different from baseline ($P = .02$). †Significantly different from baseline ($P = .03$). ‡Significantly different from baseline ($P = .05$).

incidence of injury ($P < .001$) to the thigh/calf (41.3/1000 TH, 28.1/1000 TH, 21.7/1000 TH), knee (19.3/1000 TH, 7.5/1000 TH, 7.7/1000 TH), and ankle/foot (31.9/1000 TH, 18/1000 TH, 13.2/1000 TH) were observed in semiprofessional players with peak TL during preseason (1891 AU weekly) compared with midseason (1345 AU weekly) and late season (1488 AU weekly) phases.[16] Similarly, TL over 3 consecutive 14-week preseasons (326, 260, 298 AU per session 2001, 2002, 2003, respectively) was associated with higher injury risk patterns for thigh/calf (54.1/1000 TH, 32.6/1000 TH, 17.6/1000 TH) and ankle/foot (43/1000 TH, 19.7/1000 TH, 13.5/1000 TH) (all Ps $< .001$).[17]

Three studies investigated the relationship between TL and injury mechanism (Table 5).[15–17] Overexertion was the most common cause of training injury in all 3 studies. Significant differences were observed in the incidence of overexertion (62.4/1000 TH, 22.3/1000 TH, 24.3/1000 TH) ($P < .001$) and overuse injuries (32.6/1000 TH, 18.9/1000 TH, 14.9/1000 TH) ($P < .01$) in semiprofessional players over 14 weeks between the 2001 preseason (326 AU per session) and 2002/2003 (260 and 298 AU per session, respectively) preseasons, when TL were reduced.[17] Higher incidences of overexertion (55.1/1000 TH, 22.4/1000 TH, 20.9/1000 TH) and overuse injuries (28.1/1000 TH, 18.9/1000 TH, 5.5/1000 TH) ($P = .001$) were also observed in semiprofessional players in conjunction with higher TL (1891 AU weekly, 1345 AU weekly, 1488 AU weekly pre/mid/late season, respectively).[16]

Four studies investigated the relationship between TL and injury severity (Table 6),[15,17,18,27] with 1 study identifying a

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Table 3 Summary of the Relationship Between Training Loads and Injury Type in Rugby League Players

Study	Reporting variable	Training load, AU	Injury	Strain	Sprain	Cont.	Degen.	Vertebral disc	Fracture	Cartilage	Hematoma	Overuse	Joint	Abrasion	Lacer.	Resp.	Blisters	Other	Total
Gabbett and Jenkins[18]	Noncontact	NR	Number	100.0	11.0	2.0	6.0	1.0	2.0	5.0	1.0	NR	NR	NR	NR	NR	NR	47.0	175.0
			Incidence	12.4	1.4	0.2	0.7	0.1	0.2	0.6	0.1							5.8	21.8
	Contact		Number	10.0	12.0	6.0	2.0	0.0	1.0	1.0	11.0	NR	NR	NR	NR	NR	NR	6.0	49.0
			Incidence	1.2	1.5	0.7	0.2		0.1	0.1	1.4							0.7	6.1
	Strength		Number	5.0	6.0	0.0	0.0	5.0	1.0	1.0	5.0	NR	NR	NR	NR	NR	NR	4.0	27.0
			Incidence	0.6	0.8			0.6	0.1	0.1	0.6							0.5	3.4
Gabbett[26]	2006–2007	3560.3	Number	24.0	2.0	NR	NR	NR	NR	NR	NR	4.0	NR	NR	NR	NR	NR	NR	30.0
			Incidence	1.8	0.2							0.3							2.3
	2008–2009		Number	17.0	3.0	NR	NR	NR	NR	NR	NR	6.0	NR	NR	NR	NR	NR	NR	26.0
			Incidence	1.3	0.2							0.5							2.0
Killen et al[40]	Preseason	2839.7	Number	5.0		1.0	4.0	NR	NR	NR	1.0	1.0	NR	NR	NR	NR	NR	8.0	20.0
			Incidence	1.7		0.3	1.4				0.3	0.3						2.7	6.7
Gabbett and Domrow[16]	Preseason	1891.0	Incidence	46.3*	NR	9.4	NR	NR	2.2	NR	5.5*	0.6	30.8*	17.1*	1.7	2.8	11*	10.5*	137.9
	Early season	1345.0	Incidence	32.1	NR	11.0	NR	NR	0.0	NR	1.3	0.0	13.6	7.5	0.4	0.4	5.3	3.0	74.6
	Late season	1488.0	Incidence	25.1	NR	7.2	NR	NR	0.9	NR	2.6	0.0	17.0	4.7	1.3	1.7	0.0	1.7	62.2
Gabbett[15]	Training	297.1	Number	160.0	NR	38.0	NR	NR	3.0	NR	16.0	2.0	88.0	31.0	4.0	8.0	20.0	19.0	389.0
			Incidence	43.5		10.3			0.8		4.4	0.5	24.0	8.4	1.1	2.2	5.4	5.2	105.8
	Match	212.3	Number	166.0	NR	191.0	NR	NR	18.0	NR	135.0	0.0	155.0	170.0	32.0	14.0	9.0	58.0	948.0
			Incidence	160.6		184.8			17.4		130.6		150.0	164.5	31.0	13.5	8.7	56.1	917.2
Gabbett[15,17]	2001	326.0	Number	90**	50**	18.0	NR	NR	2.0	NR	11**	NR	NR	18.0	1.0	NR	20.0	16.0	226.0
			Incidence	62.4**	43.7**	12.5			1.4		7.6**			12.5	0.7		13.9	11.1	165.8
	2002	260.0	Number	27.0	17.0	18.0	NR	NR	1.0	NR	4.0	NR	NR	24.0	1.0	NR	7.0	4.0	103.0
			Incidence	23.2	14.6	25.4			0.9		3.4			20.6	0.9		6.0	3.4	92.4
	2003	298.0	Number	29.0	22.0	16.0	NR	NR	3.0	NR	2.0	NR	NR	20.0	3.0	NR	11.0	10.0	116.0
			Incidence	19.6	14.9	10.8			2.0		1.4			13.5	2.0		7.4	6.8	78.4

Abbreviations: AU, arbitrary units; Cont., contusion; Degen., degenerative; Incidence, per 1000 training/playing hours; Lacer., laceration; NR, not reported; Number, total number of occurrences; Resp., respiratory.
*Significant difference between preseason and during competition ($P = .05$). **Significant difference between 2001 and 2002/2003 seasons ($P = .05$).

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Table 4 Summary of the Relationship Between Training Loads and Injury Site in Rugby League Players

Study	Reporting variable	Training load, AU	Injury	Face/head neck	Shoulder	Arm/hand	Thorax/ abdomen	Thigh/calf	Anterior thigh	Posterior thigh	Calf	Ankle/foot	Knee	Other
Gabbett and Jenkins[18]	Contact	NR	Number	5.0	7.0	5.0	6.0	NR	8.0	6.0	2.0	6.0	0.0	2.0
			Incidence	0.6	0.9	0.6	0.7		1.0	0.7	0.2	0.7		0.2
	Noncontact		Number	1.0	5.0	1.0	13.0	NR	28.0	38.0	32.0	20.0	23.0	14.0
			Incidence	0.1	0.6	0.1	1.6		3.5	4.7	4.0	2.5	2.9	1.7
	Strength		Number	2.0	5.0	0.0	13.0	NR	2.0	2.0	0.0	0.0	2.0	1.0
			Incidence	0.3	0.6		1.7		0.3	0.3			0.3	0.1
Gabbett[26]	2006–2007	3560.3	Number	NR	NR	NR	6.0	NR	6.0	12.0	0.0	NR	6.0	0.0
			Incidence				0.5		0.5	0.9			0.5	
	2008–2009		Number	NR	NR	NR	2.0	NR	0.0	10.0	3.0	NR	9.0	2.0
			Incidence				0.2			0.8	0.2		0.7	0.2
Killen et al[40]	Preseason	2839.7	Number	NR	1.0	1.0	2.0	7.0	NR	NR	NR	3.0	3.0	3.0
			Incidence		0.3	0.3	0.7	2.4				1.0	1.0	1.0
Gabbett and Domrow[16]	Preseason	1891.0	Incidence	10.0	2.8	11*	19.8*	41.3*	NR	NR	NR	31.9*	19.3*	1.7*
	Early season	1345.0	Incidence	7.5	2.2	4.4	8.4	28.1	NR	NR	NR	18.0	7.5	0.0
	Late season	1488.0	Incidence	4.7	3.0	3.8	8.5	21.7	NR	NR	NR	13.2	7.7	0.0
Gabbett[15,17]	Training	297.1	Number	23.0	13.0	21.0	49.0	142.0	NR	NR	NR	93.0	47.0	1.0
			Incidence	6.2	3.5	5.7	13.3	38.6				25.3	12.8	0.3
	Match	212.3	Number	224.0	90.0	122.0	100.0	180.0	NR	NR	NR	91.0	131.0	10.0
			Incidence	216.7	87.1	118.0	96.8	174.2				88.1	126.8	9.7
Gabbett[15,17]	2001	326.0	Number	14.0	6.0	10.0	30**	78**	NR	NR	NR	62**	25.0	1.0
			Incidence	9.7	4.2	6.9	20.8**	54.1**				43**	17.3	0.7
	2002	260.0	Number	16.0	2.0	9.0	5.0	38.0	NR	NR	NR	23.0	14.0	3.0
			Incidence	13.7	1.7	7.7	4.3	32.6				19.7	12.0	2.6
	2003	298.0	Number	17.0	2.0	16.0	16.0	26.0	NR	NR	NR	20.0	17.0	2.0
			Incidence	11.5	1.4	10.8	10.8	17.6				13.5	11.5	1.4

Abbreviations: AU, arbitrary units; Incidence, per 1000 training/playing hours; NR, not reported; Number, total number of occurrences.
*Significant difference between preseason and during competition ($P = .05$). **Significant difference between 2001 and 2002/2003 seasons ($P = .05$).

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Table 5 Summary of the Relationship Between Training Loads and Injury Mechanism in Rugby League Players

Study	Reporting variable	Training load, AU	Injury	Overuse	Overexertion	Being tackled	Tackling	Struck by opposition	Collision with object/ player	Ground contact	Fall/slip/ trip	Struck by ball	Other
Gabbett and Domrow[16]	Preseason	1891.0	Incidence	28.1*	55.1*	5*	2.8	NR	14.9	14.3*	11.6	NR	6.1
	Early season	1345.0	Incidence	18.9	22.4	1.3	0.4	NR	16.7	4.4	8.8	NR	3.1
	Late season	1488.0	Incidence	5.5	20.9	0.9	1.3	NR	14.9	4.7	10.6	NR	3.8
Gabbett[15]	Training	297.1	Number	72.0	137.0	3.0	2.0	12.0	81.0	NR	52.0	5.0	25.0
			Incidence	19.6	37.3	0.8	0.5	3.3	22.1		14.2	1.4	256.8
	Match	212.3	Number	19.0	75.0	209.0	153.0	159.0	168.0	NR	66.0	2.0	91.0
			Incidence	18.4	41.6	202.2	148.0	153.8	162.6		63.9	1.9	88.1
Gabbett[15,17]	2001	326.0	Number	47**	90**	2**	1.0	NR	31.0	16.0	27**	NR	12**
			Incidence	32.6**	62.4**	1.4**	0.7		21.5	11.1	18.7**		8.3**
	2002	260.0	Number	22.0	26.0	1.0	2.0	NR	27.0	16.0	11.0	NR	5.0
			Incidence	18.9	22.3	0.9	1.7		23.2	13.7	9.4		4.3
	2003	298.0	Number	22.0	36.0	8.0	4/2.7	NR	24.0	13.0	8.0	NR	1.0
			Incidence	14.9	24.3	5.4	2.7		16.2	8.8	5.4		0.7

Abbreviations: AU, arbitrary units; Incidence, per 1000 training/playing hours; NR, not reported; Number, total number of occurrences
*Significant difference between preseason and during competition ($P = .05$). **Significant difference between 2001 and 2002/2003 seasons ($P = .05$).

Table 6 Summary of Relationship Between Training Loads and Injury Severity

Study	Reporting variable	Training load, AU	Injury	Transient		Minor		Moderate		Major		Total	
				Match	Training	Match	Training	Match	Training	Match	Training	Match	Training
Gabbett and Jenkins[18]	Noncontact	NR	Number	154.0	98.0	8.0	48.0	9.0	18.0	4.0	11.0	21.0	77.0
			Incidence	19.1	12.2	1.0	6.0	1.1	2.2	0.5	1.4	2.6	9.6
	Contact	NR	Number	41.0	20.0	3.0	16.0	5.0	11.0	0.0	2.0	8.0	29.0
			Incidence	5.1	2.5	0.4	2.0	0.6	1.4		0.2	1.0	3.6
	Strength	NR	Number	NA	15.0	NA	8.0	NA	4/0.5	NA	0.0	NA	27.0
			Incidence		1.9		1.0						1.5
Gabbett[15]	Training	297.1	Number	881.0	381.0	27.0	4.0	30.0	4.0	10.0	0.0	948.0	389.0
	Match	212.3	Incidence	852.4	103.7	26.1	1.1	29.0	1.1	9.7		917.3	105.9
Gabbett[15,17]	2001	326.0	Number	NR	178*	NR	44*	NR	3.0	NR	1.0	NR	226.0
			Incidence		123.4*		30.5*		2.1		0.7		1442.4
	2002	260.0	Number	NR	75.0	NR	33.0	NR	2.0	NR	0.0	NR	110.0
			Incidence		64.3		28.3		1.7				1165.9
	2003	298.0	Number	NR	88.0	NR	22.0	NR	5.0	NR	1.0	NR	116.0
			Incidence		59.5		14.9		3.4		0.7		1478.9

Abbreviations: AU, arbitrary units; incidence, per 1000 training/playing hours; NA, not applicable; NR, not reported; number, total number of occurrences.
*Significant difference between 2001 and 2002/2003 seasons ($P = .05$).

relationship.[29] The majority of training injuries (97.9%) in 2 studies were reported as transient.[15,17] Significantly, the incidence of more severe injuries (0.7/1000 TH vs 0.0/1000 TH, respectively) was reported in semiprofessional players in the 2001 preseason when TL were higher (326 AU per session), compared with the 2002 (260 AU per session) preseason ($P < .05$).[17] Interestingly, in 1 study of elite players, noncontact injuries (77) were almost 3 times higher than contact injuries (29) resulting in lost training or game time.[18]

Discussion

To our knowledge, this is the first systematic review examining the available evidence surrounding TL of rugby league players and its impact on physical performance and injury. The findings indicate that (1) TL was associated with increases in maximal aerobic power, lower body muscular power, and agility; (2) higher TL was associated with higher incidences of muscular strains and joint sprains; (3) higher TL was associated with increased incidence of injury to the thigh/calf, knee, and ankle/foot; (4) higher TL was associated with higher incidences of overexertion and overuse injuries; and (5) consistent flaws in study design exist within the current literature base which include but are not limited to issues surrounding external validity, bias, and confounding.

Findings from 2[16,17] of 3 studies that tracked TL intraseason or interseason in subelite populations indicate a dose–response effect on both physical performance and injury incidence characteristics in rugby league players. Improvements in physical performance measures of maximal aerobic power, lower body muscular power, and agility are clearly demonstrated in response to twice-weekly sessions over an entire season (9 mo) or preseason (14 wk), with session TL ranging from 260 to 469.7 AU. While familiarity of task must always be considered in relation to performance measures such as these, nonetheless, improvements here were statistically significant.

Equally demonstrable were the negative effects of TL on injury. Higher TL was associated with higher training injury incidence, more muscular strains, injuries to lower limbs, and overexertion injuries. These injury characteristics are consistent with the mechanics of running and thus higher rugby league TL.

Maximal aerobic power, lower body muscular power, and agility increased during the preseason training period when TL was at its highest (1891 vs 1416.5 AU weekly). Maximal aerobic power was maintained throughout the length of the competitive season, indicating in-season TL (average session TL = 202.7 AU), although substantially lower than preseason TL (average session TL = 275.9 AU), in conjunction with match intensity, was adequate enough to maintain aerobic performance improvements. However, increases in lower body power and agility were not maintained throughout the competitive season, suggesting variable, inadequate TL or other factors (eg, competition fatigue) may also be associated with declines in some physiological parameters during the competition phase. It is possible that a minimum amount of TL which targets lower body power and agility may be required to maintain accrued adaptations.

A relationship between TL and injury incidence was evident in 3 identified studies examining players across and within competitive seasons. Within a single season increased injury risk was evident during preseason when average session and/or weekly TL was highest. Thus, if TL is highest early in preseason and a player has been exposed to little or no TL in the off-season, higher injury incidence is likely. This is especially true for less experienced or younger populations. In addition, when TL was reduced over consecutive seasons, reductions in injury incidence were evident. However, incidence remained elevated early in the preseason.

Analysis of injury characteristics suggest that certain soft-tissue injury types (ie, strains and sprains) may be sensitive to fluctuations in TL, as reductions in TL were associated with concomitant reduction in these types of injuries in semiprofessional players. By contrast, no such relationship was identified for injuries which are typically incurred via contact mechanisms (ie, fractures, lacerations), suggesting these injury types occur independent of TL. Relationships between TL and injury site were observed only for common lower limb areas (eg, thigh, knee, calf, and ankle). Higher TL was also associated with higher incidences of overexertion and overuse injury mechanisms. These types, sites, and mechanisms of injuries typically correlate with overall running loads. The results suggest that reductions in TL may lead to corresponding reductions in muscular strains to the lower limb region resulting from overuse or overexertion mechanisms.

A recently proposed acute:chronic workload ratio model has been presented in a recent series of rugby league studies.[8,38,39] The ratio suggests increased injury risk occurs when acute TL (for a given week) exceeds chronic loads (previous 4 wk cumulative TL) with a z score >1 in elite players. The model indicates an ideal ratio range of 0.8 to 1.3, where injury risk is at its lowest. The model was initially applied in a retrospective study on cricket fast bowlers.[45] The premise proposed in the rugby league model[8] suggests that substantial increases (>10%) in weekly TL are associated with increased injury incidence, while simultaneously elucidating protective benefits from onward engagement in higher chronic TL (>16,000 m). Here, it has been shown that, with adequate chronic TL, athletes can tolerate up to 30% acute increases in weekly TL. Two studies have since applied the model to elite-level rugby league players, using GPS to measure TL over consecutive seasons.[38,39] However, rugby league players' training demands are multifaceted (consisting of skills, aerobic/anaerobic conditioning, strength, speed, agility, and power training), so quantifying an accurate estimate of overall TL requires multifaceted assessment (ie, accurate measures of training subcomponents). Many of these modalities (skills, aerobic/anaerobic conditioning, speed, and agility) can be accurately measured using GPS technology, discrete movement velocities, acceleration bands, and associated heart rate data. Hypothetically, this would help establish more informed training programs and longitudinal application of TL according to the developmental stage (eg, junior to senior professional) and individual needs of players (eg, rehabilitation from injury). Recently, a model for systematic monitoring and progression of TL aimed at maximizing overall athletic performance and minimizing negative training outcomes throughout the various stages of athlete development was proposed.[46]

A number of key limitations can be identified from this systematic review. First, there are limited numbers of independent studies in total, suggesting that the current evidence base on TL assessment and the relationships with physical performance and injury is restricted. Furthermore, the methodological assessment here identified several key recurring flaws in the study design (studies score poorly on subcategories of external validity, bias, and confounding). Issues of external validity could be improved with identification of recruiting strategies and screening criteria, while issues of potential bias could have been partially accounted for with discussion of participant compliance, dropout, and follow-up. In addition, the inclusion of a control group for all studies would have enhanced the study quality greatly.

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Second, there is no standardized definition of TL or uniform quantification of overall TL. In 9 of our 12 identified studies, only a single method of quantifying TL based on internal perceived TL was utilized, and minimal data exist using external objective TL measurement (ie, GPS technology). This is likely due to most of the studies here predating the widespread use of GPS technology in rugby league. In addition, most studies included in this review were conducted in the subelite populations, who typically do not have access to GPS devices. As critical to TL definitions and measurement is the need for specific detail and consistent reporting of TL accumulation. For example, TL was variably reported in studies as a measure of total distance run (n = 3), total weekly measure (in AU; n = 3), and as an average training session load (in AU; n = 4). While 10 studies indicated the number of training sessions per week, 8 did not provide the total durations (total number of weeks). Third, the challenges in accurately assessing TL are exacerbated by the lack of assessment of additional training modalities composing rugby league training. For instance, to date only 1 study has examined the relationship between resistance training–based TL and injury incidence. Here, strength and power were significantly related to injuries sustained during strength/power training activities ($r = .63$, $P < .05$) as well as on-field contact ($r = .75$, $P < .01$) and noncontact ($r = .82$, $P < .01$) injuries.[18] Thus, considering and capturing TL across facets of training could be important. Existing absolute (volume load) and allometric scaling methods (volume index)[47] used in strength and power sports could be integrated to help estimate overall total TL assessments. Fourth, the included studies do not include a homogeneous cohort. As a result, the variability in participant characteristics (stages of development, skill sets, and experiences) do not allow for comparisons across studies.

Future studies would benefit from a more conclusive definition of overall TL that include accurate, consistent quantification of both internal and external measures. Detailed reporting of training programs and how accumulated TL is calculated (ie, per session, week, or longer) is required. This will help achieve standardized calculation of overall accumulated TL. Assessment of training subcomponents would permit an examination as to how other forms of training modalities could be related to both performance measures and injury incidence. Finally, longitudinal studies conducted across several tiers of playing groups simultaneously would provide valuable insight into the dynamic nature of TL and its relationships with performance and injury. That said, multiple challenges exist in attempting to implement more comprehensive study methodologies. These include but are not limited to resources (equipment such as GPS for measurement of external loads is expensive), participation and compliance (access to elite-level populations is difficult to come by due to the nature of elite-level sport, while compliance is a key issue with athletes competing at lower levels), and the fact that integration of data obtained from different training methodologies is a complex issue that is, at present, difficult to address.

Conclusions

The purpose of this review was to investigate the relationships between TL, physical performance measures, and injury incidence. On the basis of existing evidence, there appears to be an association between TL and measures of maximal aerobic power, muscular power, and agility. Similarly, relationships between TL and injury characteristics also appear apparent, notably overall injury incidence, type, site, and mechanism. Higher TL is associated with the occurrence of specific soft-tissue injury types, and lower limb injuries via overuse and overexertion mechanisms. The limitations evident in existing studies make it challenging to identify the exact nature of these relationships and the factors that attenuate them. While acknowledging limitations based on accurate standardized measures of TL in rugby league training modalities, the findings suggest value in the development of a model to maximize overall athletic performance and minimize negative training outcomes. Such a model could have important implications for TL management in elite and player development systems.

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